Individual Contribution Report

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Group 35

1. Tasks & Contributions

My contribution is in the core recovery algorithm and the rules that guarantee: committed updates are retained; uncommitted updates are undone. Specifically,

- **WAL + log record design:** Designed the logging mechanism, the log record structure and what information is recorded to help perform recovery (transaction history and update record).
- **Recovery workflow:** Implemented the process to recover on system startup using the basic process of analysis, followed by redo and undo.
- **Checkpointing**: Implemented the code that helps perform recovery in an effective manner while recovering from checkpoints.
- **Debugging & verification:** Had various types of crashes (mid-update, after few log entries and before/after checkpoint point). helpful in identifying and correcting any bugs in my implementation of incorrect redo/undo decisions. I also added some documentation to the recovery parts of the code.

2. Collaboration & Challenges

Coordinated using Git, we divided our tasks into separate modules. The most difficult aspect was creating scenarios where everything worked properly, but then crashed and stopped functioning as expected. Our process involved creating artificial crashes, seeing what can persist and ensuring that we achieve the desired state after recovery.

3. Self-Assessment

With this project, I gained practical knowledge regarding recovery techniques, i.e., significance of WAL ordering, nature of changes in checkpoints, relationships among Analysis, Redo, and Undo. I believe that the most significant aspect of my contribution lies in ensuring the end-to-end correctness of recovery techniques. Had I got another chance to undertake this project, I would include more automated crash tests.

4. Use of AI Tools

I used AI Tools (ChatGPT, Claude) for:

- Concept clarification: I used it to confirm ARIES/WAL concepts (what must be logged, what each phase does in Analysis / Redo / Undo, and how checkpointing affects recovery).
- Debugging support: When recovery output didn’t match expectations (e.g., committed vs. uncommitted effects after a crash), I used AI to brainstorm possible causes and edge cases to test.
- Documentation help: I used AI to polish short explanations/comments after I finalized the implementation, so the wording is clearer and consistent.